“I Can’t See”

Custom Object Detection

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Project Idea and Purpose

- Goal: Assist visually impaired individuals in locating everyday household objects
- Develop a real-time object detection system capable of detecting 8 everyday objects from a live camera feed
- Deploy and optimize the YOLOv8 model on NVIDIA Jetson Orin Nano
- Evaluate performance in real-world home conditions

Data Collection

- Collected custom dataset using Logitech C270 USB camera under real-world home conditions
- Focused on 8 common household objects that visually impaired users frequently need to locate:
    - **Backpack, Bottle, Cell Phone, Cup, Keyboard, Mouse, Potted Plant, Spoon**
- Total of ~1,494 labeled object instances across all classes
- Captured realistic home scenarios with variations in lighting conditions, object orientations, distances from camera, and background environments

Data Labeling

- Manually annotated images using Roboflow and LabelImg tools
- Created bounding boxes for each of the 8 object categories
- Each image labeled to identify object location and class identity
- Dataset split into training, validation, and testing sets

Training and Fine Tuning

- Model: YOLOv8 framework
- Epochs: 50
- Batch Size: 16
- Image Resolution: 640×640 pixels
- Platform: GPU training, then deployed to Jetson Orin Nano

70% Training

15% Validation

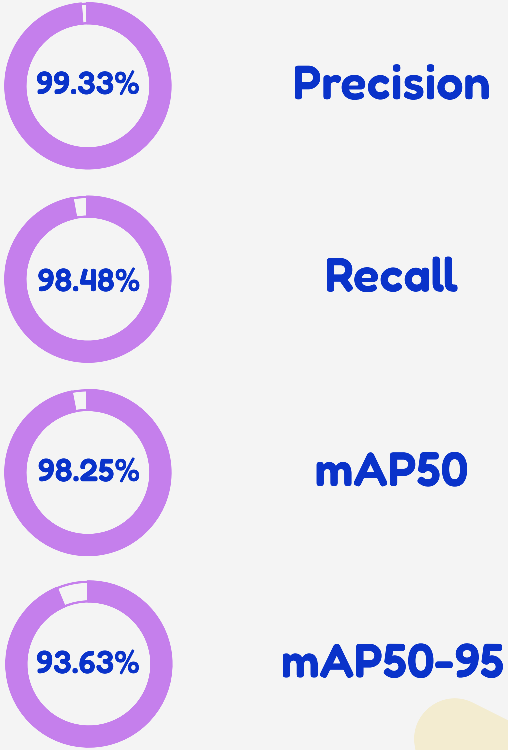
15% Testing

Training Results

Metric	Epoch 1	Epoch 50	Improvement
train/box_loss	1.1293	0.2835	75% reduction
train/cls_loss	2.0954	0.2121	90% reduction
train/dfl_loss	1.2848	0.8652	33% reduction

Testing - Detection Performance

- **Precision: 99.33%** - When model predicts an object, it's correct 99.33% of the time
- **Recall: 98.48%** - Model successfully detects 98.48% of all objects present
- **mAP50: 98.25%** - Mean Average Precision at 50% IoU threshold
- **mAP50-95: 93.63%** - Mean Average Precision across IoU thresholds from 50% to 95%
- Near-perfect detection accuracy critical for assisting visually impaired users
- Model performs excellently on unseen data



Deployment

- Deployed the optimized YOLOv8 model on the NVIDIA Jetson Orin Nano
- Optimized model using tensorRT
- Connected Logitech C270 USB camera for live video input
- Real-time object detection processing:

    Camera captures frames at 640×640 resolution

    Jetson GPU accelerates inference

    Bounding boxes and class labels displayed in real-time

- Portable edge device suitable for wearable or mobile assistive applications
- System achieves near real-time performance on embedded hardware

Future Implementations

Text-to-Speech Integration: Implement audio output to announce detected objects out loud

- Example: "Backpack detected in front of you"
- Real-time audio feedback for hands-free operation

Expand Object Classes: Add more essential household items (keys, wallet, glasses, remote control, phone charger)

Spatial Audio Guidance: Provide directional information

- "Cell Phone detected to your left"
- Distance estimation: "Bottle is 3 feet away"

Optimize for Mobile Devices: Deploy on smartphones for broader accessibility

Thank You!

Do you have any questions?